

Today's Topic : Data Compression

encoding with fewer bits.

$$(x_1, \dots, x_n) \in \mathcal{X}^n$$

$\mathcal{X}$: event space.

Goal : How many bits are needed to indicate whole seqs ?

ex) $\mathcal{X} = \{1, \dots, 10\}$. $\mathcal{X}^3$

$$\hookrightarrow 2^p < 10^3 < 2^{10} \Rightarrow 10 \text{ bits? } \times$$

if $P(X=5)=1$ ($\Rightarrow P(X \neq 5)=0$),

$\Rightarrow (5, 5, 5) \Rightarrow 1 \text{ bit!}$

Probability Works!

Two Steps. $\rightarrow$ "typical set".

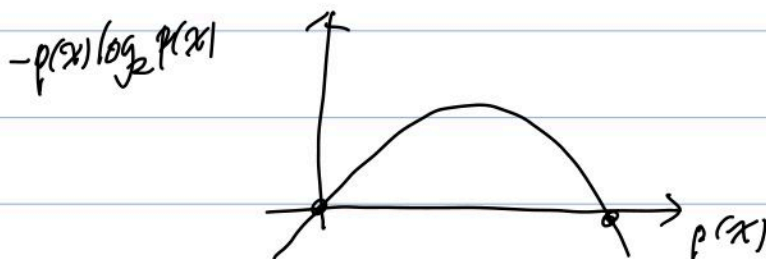
[1] Select Target Seq.

[2] Count.

Review

X : random variable $\sim p(x)$. (H)

The "Entropy" of X : $H(X) := - \sum_x p(x) \log_2 p(x)$.



$(X_1, \dots, X_n)$

(H) (T)

independent, identically distributed (i.i.d)

Chapter 3. Asymptotic Equipartition Property.

§3.1. Asymptotic Equipartition Property. thm.

The Law of Large Numbers

For i.i.d r.v X ,

expected value of X

$$\frac{1}{n} \sum_{i=1}^n X_i \longrightarrow \underline{E(X)} \quad \text{as } n \rightarrow \infty.$$

Asymptotic Equipartition Theorem (AEP)

For i.i.d r.v $X_1, \dots, X_n$ and $p(x)$,

$$\frac{1}{n} \log_2 \frac{1}{p(X_1, \dots, X_n)} \rightarrow H(X) \quad \text{as } n \rightarrow \infty.$$

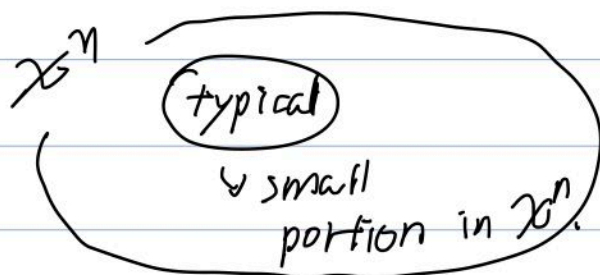
$$\therefore p(X_1, \dots, X_n) \rightarrow 2^{-nH} \quad \text{as } n \rightarrow \infty.$$

$$\begin{aligned} \langle \text{pf} \rangle. \quad \frac{1}{n} \log_2 \frac{1}{p(X_1, \dots, X_n)} &= \frac{1}{n} \sum_{i=1}^n \log \frac{1}{p(X_i)} \xrightarrow[n \rightarrow \infty]{\text{L.L.N}} E\left[\log \frac{1}{p(X)}\right] \\ &\quad \parallel \\ &H(X). \quad \square \end{aligned}$$

Def $A_\varepsilon^{(n)} := \{ (x_1, \dots, x_n) \in \mathcal{X}^n : 2^{-n(H(X)+\varepsilon)} \leq p(x_1, \dots, x_n) \leq 2^{-n(H(X)-\varepsilon)} \}$.

"Typical set"

$$\mathcal{X}^n = \underbrace{A_\varepsilon^{(n)}} \cup \underbrace{(\mathcal{X}^n \setminus A_\varepsilon^{(n)})}$$



Thm 1) If $(x_1, \dots, x_n) \in A_\epsilon^{(n)}$,

$$H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, \dots, x_n) \leq H(X) + \epsilon.$$

ii) For sufficiently large n ,

$$Pr \{ A_\epsilon^{(n)} \} > 1 - \epsilon$$

$$iii) |A_\epsilon^{(n)}| \leq 2^{n(H(X) + \epsilon)}$$

$$iv) |A_\epsilon^{(n)}| \geq (1 - \epsilon) 2^{n(H(X) - \epsilon)}$$

for sufficiently large n .

<pf> i) By def.

$$ii) \lim_{n \rightarrow \infty} Pr \{ (X_1, \dots, X_n) \in A_\epsilon^{(n)} \} = 1.$$

$$\delta = \epsilon$$

$$\forall \delta > 0, \exists n_0 \in \mathbb{N} \text{ s.t.}$$

$$Pr \{ | -\frac{1}{n} \log p(X_1, \dots, X_n) - H(X) | < \epsilon \} > 1 - \delta.$$

$$iii) 1 = \sum_{x \in \mathcal{X}^n} p(x) \geq \sum_{x \in A_\epsilon^{(n)}} p(x) \geq \sum_{x \in A_\epsilon^{(n)}} 2^{-n(H(X) + \epsilon)} = |A_\epsilon^{(n)}| \cdot 2^{-n(H(X) + \epsilon)}.$$

iv) From ii), $Pr \{ A_\epsilon^{(n)} \} > 1 - \epsilon$ - suff. large n .

$$1 - \epsilon < Pr \{ A_\epsilon^{(n)} \} \leq \sum_{x \in A_\epsilon^{(n)}} 2^{-n(H(X) - \epsilon)} = 2^{-n(H(X) - \epsilon)} |A_\epsilon^{(n)}|. \quad \square$$

3.2. Consequences of AEP: Data Compression.

ex) $|A_\epsilon^{(n)}| = 4 = 2^2$

	0.	1.	2.	3.	
(Bin)	↓	↓	↓	↓	
	<u>00</u>	<u>01</u>	<u>10</u>	<u>11</u>	$\Rightarrow 2 \text{ bits needed.}$

⊙ Typical Set.

$$|A_\epsilon^{(n)}| = 2^{\log_2 |A_\epsilon^{(n)}|} \Rightarrow \exists k \in \mathbb{N} \text{ s.t. } k < \log_2 |A_\epsilon^{(n)}| \leq k+1.$$

$$\Rightarrow k+1 \text{ bits. } ? \in \mathbb{N}?$$

$$\text{Since } |A_\epsilon^{(n)}| \leq 2^{n(H(X) + \epsilon)}, \quad \log_2 |A_\epsilon^{(n)}| \leq \underline{n(H(X) + \epsilon)}.$$

$$\Rightarrow \text{Need } n(H(X) + \epsilon) + 1 \text{ bits!}$$

To indicate the elt "in $A_\epsilon^{(n)}$ ", start with 0.

i.e. $0 \underbrace{\quad \quad \quad}_{n(H+\epsilon)+1} \Rightarrow$ need $n(H(X)+\epsilon)+2$ for indexing.

② Non-Typical Set.

$$|\mathcal{X}^n| = 2^{\log_2 |\mathcal{X}|^n} \Rightarrow \log_2 |\mathcal{X}|^n + 1 \text{ bits.}$$

To indicate it's "in $(A_\epsilon^{(n)})^c$ ", start with 1!

i.e. $1 \underbrace{\quad \quad \quad}_{\log_2 |\mathcal{X}|^n + 1} \Rightarrow$ need $n \log_2 |\mathcal{X}| + 2$ bits!

$$X^n := (x_1, \dots, x_n)$$

$l(X^n) :=$ length of the codeword corresp. to X^n .

Thm $X^n: \text{i.i.d. } \sim p(x), \epsilon > 0$

Then, $\exists$ a code that maps X^n into binary string
s.t. mapping is 1-1 ($\Rightarrow$ invertible) and

$$E\left[\frac{1}{n} l(X^n)\right] \leq H(X) + \epsilon$$

for suff. large n .

<pf>. Assume n is suff. large. $\Rightarrow \Pr\{X^n \in A_\epsilon^{(n)}\} > 1 - \epsilon$.

$$E(l(X^n)) = \sum_{x^n \in \mathcal{X}^n} p(x^n) l(x^n) = \sum_{x^n \in A_\epsilon^{(n)}} p(x^n) l(x^n) + \sum_{x^n \in A_\epsilon^{(n)c}} p(x^n) l(x^n)$$

$$= \sum_{x^n \in A_\epsilon^{(n)}} p(x^n) \cdot (n(H+\epsilon)+2) + \sum_{x^n \in A_\epsilon^{(n)c}} p(x^n) (n \log |\mathcal{X}| + 2)$$

$$= \Pr\{X^n \in A_\epsilon^{(n)}\} (n(H+\epsilon)+2) + \Pr\{X^n \in A_\epsilon^{(n)c}\} (n \log |\mathcal{X}| + 2)$$

$$\leq (1-\epsilon) (n(H+\epsilon)+2) + \epsilon (n \log |\mathcal{X}| + 2)$$

$$\leq n(H+\epsilon) + \epsilon n \log |\mathcal{X}| + 2 \stackrel{\uparrow}{=} n(H+\epsilon')$$

$$\varepsilon' = \varepsilon + \varepsilon \log |\mathcal{X}| + \frac{2}{n}.$$

$$\varepsilon \rightarrow 0, n \rightarrow \infty \Rightarrow \varepsilon' \rightarrow 0 \quad \square$$

3.3. High-Probability Sets and the Typical Sets.

Def Let $B_\delta^{(n)} \subset \mathcal{X}^n$ be the smallest set s.t.
 $\Pr\{B_\delta^{(n)}\} \geq 1 - \delta$. $\rightarrow$ high-probability set.

Thm $X_1, \dots, X_n$ i.i.d. $\sim p(\mathcal{X})$. For $\delta < \frac{1}{2}$ and $\delta' > 0$,
 if $\Pr\{B_\delta^{(n)}\} > 1 - \delta$, then for suff. large n ,

$$\frac{1}{n} \log_2 |B_\delta^{(n)}| > H - \delta'.$$

Bmk. $|B_\delta^{(n)}| \geq 2^{nH}$ and $|A_\varepsilon^{(n)}| \cong 2^{n(H \pm \varepsilon)}$.
 $\Rightarrow |A_\varepsilon^{(n)}| \cong |B_\delta^{(n)}|$.

Def. $a_n \stackrel{\circ}{=} b_n \iff \lim_{n \rightarrow \infty} \frac{1}{n} \log \frac{a_n}{b_n} = 0$.

Bmk $a_n \stackrel{\circ}{=} b_n$ means a_n & b_n are equal to the first order in the exponent.

$$\langle p.p. \rangle \lim_{n \rightarrow \infty} \frac{1}{n} \log \frac{a_n}{b_n} = 0 \iff \log \frac{a_n}{b_n} = o(n)$$

$$\iff \frac{a_n}{b_n} = 2^{o(n)} \iff a_n = 2^{o(n)} \cdot b_n.$$

$\square$